Charlie Wu

charliewu.math@gmail.com • cwu4570.github.io

Current Position

2026–present **Postdoctoral Fellow**, Max Planck Institute for Mathematics
Bonn, Germany

Education

2021–2026 **PhD in Mathematics**, University of Toronto
Toronto, Ontario
Advisor: Daniel Litt
Thesis: *Character Varieties and Hodge Theory*

Research Interests

Algebraic geometry and number theory.

Publications and Preprints

2026 **Privileged SL_3-local systems are motivic**
Joint with Julia Meng
arXiv:2610.17173

2025 **Hypergeometric local systems and parabolic bundles**
arXiv:2509.25629

2025 **Minimal energy local systems on curves**
Accepted at *Épijournal de Géométrie Algébrique*
arXiv:2503.19250

Talks

2026 *Rank 3 privileged local systems* (joint talk with Julia Meng)
A Sprint on Local Systems, University of Wuppertal

2026 *Introduction to Higgs bundles*
Geometric Correspondences in the Desert, Anza-Borrego Desert State Park

2026 *Compact components of character varieties*
Algebraic Geometry Seminar, Johns Hopkins University

2026 *Compact components of character varieties*
Algebraic Geometry Seminar, University of Chicago

2025 *Compactness and character varieties*
Ottawa/Carleton Number Theory Seminar, University of Ottawa

2025 *Minimal energy local systems on curves*
Algebraic Geometry Seminar, University of Toronto

2024 *Non-abelian Hodge theory and complex variations of Hodge structure*
Elden Elmanto's Summer Seminar, University of Toronto

2024 *Six functor formalisms*
Étale Cohomology Seminar, University of Toronto

2024 *Intro to Higgs bundles and non-abelian Hodge theory*
Non-Abelian Hodge Theory Learning Seminar, University of Toronto

2024 *Étale fundamental groups*
Étale Cohomology Seminar, University of Toronto

2023 *Special unitary local systems on the Riemann sphere*
Lightning Talk, Monodromy and Its Applications Conference (Nick Katz's 80th Birthday Conference), Princeton University

2023 *Why an engineer can draw better than you*
Graduate Student Seminar, University of Toronto

2023 *Counting curves and Gromov-Witten invariants*
Graduate Student Algebraic Geometry Seminar, University of Toronto

2023 *Character varieties*
Rigid Local Systems Seminar, University of Toronto

Awards

2025 Margaret Isobel Elliott Graduate Scholarship

2018 & 2020 Drew Thompson Scholarships

Teaching Experience

University of Toronto

2026 **MAT1301** — Algebraic Topology

2025 **MAT138** — Intro to Proofs

2023 **MAT224** — Linear Algebra II

2023 **MAT237** — Multivariable Calculus with Proofs

2022–2025 **MAT334** — Complex Variables

2022 **MAT137** — Calculus with Proofs

2022 **APM346** — Partial Differential Equations

2021 **MAT344** — Introduction to Combinatorics

2019–2022 **MAT223** — Linear Algebra I

2019 **MAT136** — Calculus 1B

2018, 2021, 2022 **MAT135** — Calculus 1A

Service

2024 **Learning Seminar Organizer**
Non-Abelian Hodge Theory

2023–2024 **Mentor to First-Year Graduate Students**
Advised on core courses, qualifying exams, finding an advisor, and related topics.

2020–2021 **Vice-President of Social Affairs**
University of Toronto Mathematics Union

Citizenship: United States